



[Outline](#) > 3. Extending the model > 3.1 Interaction (Euler's method for Systems) > Mathematical model

Mathematical model

[Bookmark this page](#)

Rainbowfish and gouramis

The problem is the following

A client of the fish farm wants an aquarium containing both rainbowfish and gouramis. The client is advised that the particular kind of gouramis he wants, can get aggressive, and that they hunt and eat rainbowfish. The client is undaunted and still wants this combination. He has a budget for 20 rainbowfish, 5 gouramis and an aquarium which has a maximum capacity for 100 rainbowfish. Is it possible to have both fish in the same aquarium without one of them going extinct?

In the next video, a model is constructed for these two populations.

Interacting populations



▶ 0:00 / 5:39

▶ 1.50x



Video ▼



Subtitles ▼

Other ▼

CC BY NC ND Some Rights Reserved

Subtitles (captions) in other languages than provided can be viewed at YouTube. Select your language in the CC-button of YouTube.

In the video the system of two differential equations has been constructed for the rainbowfish population $P(t)$ and the gourami population $G(t)$. The gouramis are predators and they prey on the rainbowfish. In the next exercise you see other types of interactions.

Exercise A

1/1 point (graded)

Keyboard Help

PROBLEM

Two populations can react in different ways. Consider the four systems of differential equations below for the species X and Y. Which kind of interaction so they model?

System of differential equations	Relation between X and Y
$\frac{dX}{dt} = f_X(t, X) - \alpha XY$ $\frac{dY}{dt} = f_Y(t, Y) - \beta XY$	X and Y compete
$\frac{dX}{dt} = f_X(t, X) - \alpha XY$ $\frac{dY}{dt} = f_Y(t, Y) + \beta XY$	X is prey for Y
$\frac{dX}{dt} = f_X(t, X) + \alpha XY$ $\frac{dY}{dt} = f_Y(t, Y) + \beta XY$	X and Y cooperate
$\frac{dX}{dt} = f_X(t, X) + \alpha XY$ $\frac{dY}{dt} = f_Y(t, Y) - \beta XY$	Y is prey for X

Submit

You have used 1 of 3 attempts.



Reset



Show Answer

FEEDBACK

✓ Correctly placed 4 items.

✓ Good work! You have correctly matched the systems of differential equations with the sentences.

✓ Your highest score is 1.0

Next, you will find a page where you will learn how you can apply Euler's Method to systems of differential equations.

© All Rights Reserved



🌐 English ▼

© 2012-2017 edX Inc. All rights reserved except where noted. EdX, Open edX and the edX and Open EdX logos are registered trademarks or trademarks of edX Inc. | 粤ICP备17044299号-2

POWERED BY
OPENedX®

